

Module 2 Lab Exercise: Tools Used in Machine Learning

Learning Objectives

By the end of this lab, you will be able to:

- Set up and navigate Jupyter Notebook, Google Colab, and VS Code environments
- Install and import essential Python libraries for machine learning
- Create and format professional documentation using Markdown
- Initialize a GitHub repository for your ML projects
- Understand the basic workflow of data science tools

Prerequisites

- Basic understanding of what machine learning is (Module 1)
- Access to internet for downloading tools and datasets
- A Google account (for Colab) or local Python installation

Part 1: Environment Setup and Tool Overview

What are the main tools we'll use in this course?

Jupyter Notebook/Google Colab: Interactive computing environments where you can write code, see results immediately, and document your work with text and visualizations.

Python Libraries: Pre-written code packages that make machine learning tasks easier:

- **Pandas:** For working with data (like Excel, but more powerful)
- **NumPy:** For mathematical operations on arrays of numbers
- **Matplotlib:** For creating charts and graphs
- **Scikit-learn:** The main library for machine learning algorithms

GitHub: A platform to store, share, and collaborate on code projects

VS Code: A powerful text editor for writing and debugging code

Let's start by setting up our environment!

✓ Environment Setup Instructions

Option 1: Google Colab (Recommended for Beginners)

1. Go to colab.research.google.com
2. Sign in with your Google account
3. Click "New Notebook"
4. You're ready to go! Libraries are pre-installed.

Option 2: Local Jupyter Notebook

1. Install Python from python.org
2. Open terminal/command prompt
3. Run: `pip install jupyter pandas numpy matplotlib scikit-learn`
4. Run: `jupyter notebook`
5. Create a new notebook

Option 3: VS Code

1. Download VS Code from code.visualstudio.com
2. Install Python extension
3. Install Jupyter extension
4. Create a new .ipynb file

For this lab, we recommend starting with Google Colab as it requires no installation.

```
Install required libraries (uncomment if needed)
!pip install pandas numpy matplotlib scikit-learn

Import libraries with standard aliases
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets
import warnings
warnings.filterwarnings('ignore') # Hide warning messages for cleaner output

print("✅ All libraries imported successfully!")
print(f"Pandas version: {pd.__version__}")
print(f"NumPy version: {np.__version__}")
```

```
✅ All libraries imported successfully!
Pandas version: 2.2.3
NumPy version: 2.1.3
```

✓ Part 2: Loading and Exploring Your First Dataset

We'll use the famous Iris dataset - a classic dataset for beginners. It contains measurements of iris flowers from three different species.

```
# Load a simple dataset (Iris flowers - a classic beginner dataset)
from sklearn.datasets import load_iris

# Load the data
iris = load_iris()
print("Dataset loaded successfully!")
print(f"Dataset shape: {iris.data.shape}")
print(f"Features: {iris.feature_names}")
print(f"Target classes: {iris.target_names}")
```

```
Dataset loaded successfully!
Dataset shape: (150, 4)
Features: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal
Target classes: ['setosa' 'versicolor' 'virginica']
```

```
# Convert to pandas DataFrame for easier handling
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target_names[iris.target]

# Display first few rows
print("First 5 rows of our dataset:")
print(df.head())

print("\nDataset info:")
print(df.info())
```

```
First 5 rows of our dataset:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm) \
0                5.1                3.5                1.4                0.2
1                4.9                3.0                1.4                0.2
2                4.7                3.2                1.3                0.2
3                4.6                3.1                1.5                0.2
4                5.0                3.6                1.4                0.2
```

```
   species
0  setosa
1  setosa
2  setosa
3  setosa
4  setosa
```

```
Dataset info:
<class 'pandas.core.frame.DataFrame'>
```

```

RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   sepal length (cm)      150 non-null   float64
1   sepal width (cm)       150 non-null   float64
2   petal length (cm)      150 non-null   float64
3   petal width (cm)       150 non-null   float64
4   species                 150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None

```

✓ Part 3: Creating Your First Visualization

Data visualization is crucial in machine learning. Let's create a simple plot to understand our data.

```

# Create a simple scatter plot
plt.figure(figsize=(10, 6))

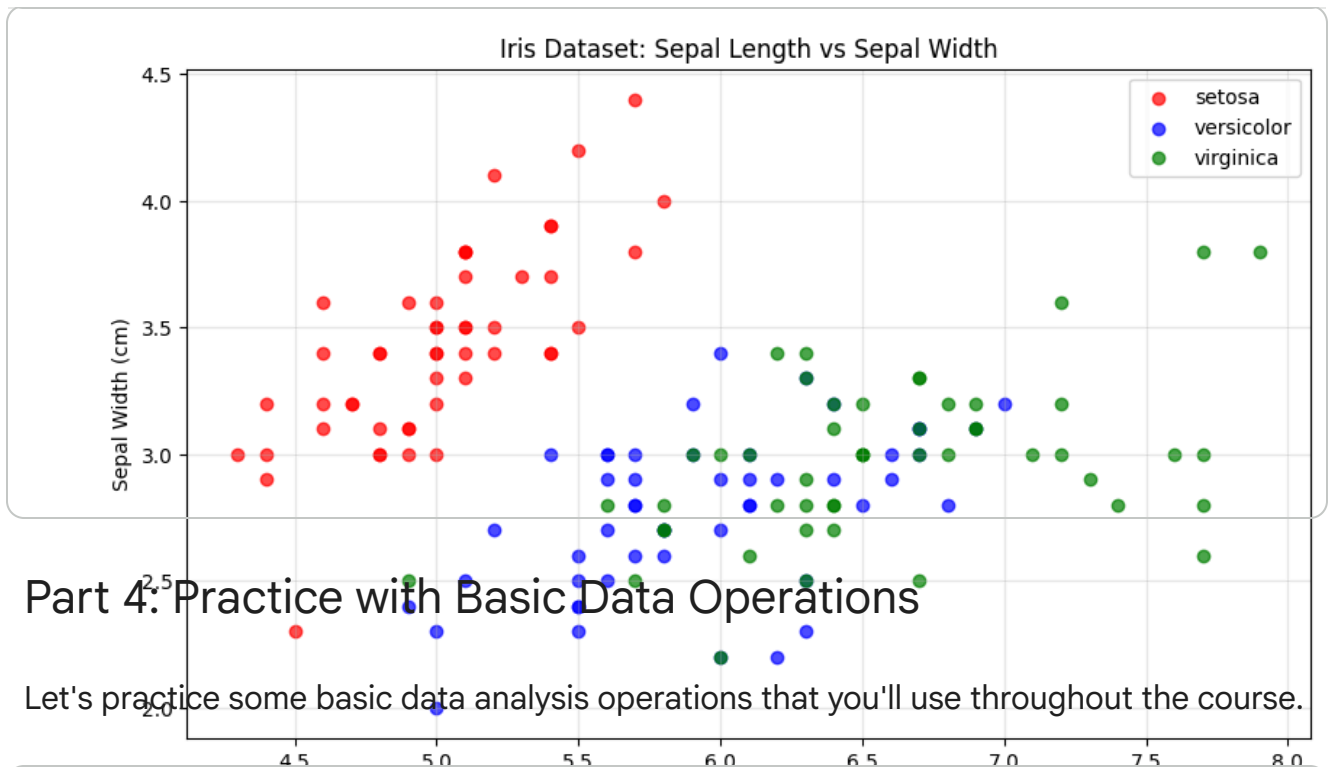
# Plot sepal length vs sepal width, colored by species
species_colors = {'setosa': 'red', 'versicolor': 'blue', 'virginica': 'green'}

for species in df['species'].unique():
    species_data = df[df['species'] == species]
    plt.scatter(species_data['sepal length (cm)'],
                species_data['sepal width (cm)'],
                c=species_colors[species],
                label=species,
                alpha=0.7)

plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')
plt.title('Iris Dataset: Sepal Length vs Sepal Width')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()

print("🎉 Congratulations! You've created your first data visualization!")

```



✓ Part 4: Practice with Basic Data Operations

Let's practice some basic data analysis operations that you'll use throughout the course.

```
# Basic statistical analysis
print("Basic Statistics for Iris Dataset:")
print("=" * 40)

# Calculate mean values for each species
species_means = df.groupby('species').mean()
print("\nMean values by species:")
print(species_means)

# Count samples per species
species_counts = df['species'].value_counts()
print("\nSamples per species:")
print(species_counts)
```

Basic Statistics for Iris Dataset:

=====

Mean values by species:

	sepal length (cm)	sepal width (cm)	petal length (cm) \
species			
setosa	5.006	3.428	1.462
versicolor	5.936	2.770	4.260
virginica	6.588	2.974	5.552

petal width (cm)

species	
setosa	0.246
versicolor	1.326
virginica	2.026

Samples per species:

species

```
setosa      50  
versicolor 50  
virginica   50  
Name: count, dtype: int64
```

Part 5: GitHub and Documentation Best Practices

Why GitHub for Machine Learning?

- **Version Control:** Track changes to your code and data
- **Collaboration:** Work with others on projects
- **Portfolio:** Showcase your work to potential employers
- **Backup:** Never lose your work

Basic GitHub Workflow:

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For This Course:

- Create a repository named "ITAI-1371-ML-Labs"
- Upload each lab notebook as you complete it
- Include a README.md file describing your projects

Action Item: After this lab, create your GitHub account and repository.

✓ Assessment: Tool Familiarity Check

Complete the following tasks to demonstrate your understanding of the tools:

```
# Task 1: Create a simple calculation using NumPy  
# Calculate the mean and standard deviation of sepal length  
  
sepal_lengths = df['sepal length (cm)']  
  
# Your code here:  
mean_sepal_length = np.mean(sepal_lengths)  
std_sepal_length = np.std(sepal_lengths)  
  
print(f"Mean sepal length: {mean_sepal_length:.2f} cm")
```

```
print(f"Standard deviation: {std_sepal_length:.2f} cm")
```

```
# Verification (don't modify)
```

```
assert isinstance(mean_sepal_length, (float, np.floating)), "Mean should be a r  
assert isinstance(std_sepal_length, (float, np.floating)), "Std should be a num  
print("✅ Task 1 completed successfully!")
```

Mean sepal length: 5.84 cm

Standard deviation: 0.83 cm

✅ Task 1 completed successfully!

```
# Task 2: Create a simple bar chart showing species counts
```

```
species_counts = df['species'].value_counts()
```

```
plt.figure(figsize=(8, 5))
```

```
plt.bar(species_counts.index, species_counts.values, color=['red', 'blue', 'green'])
```

```
plt.title('Number of Samples per Species')
```

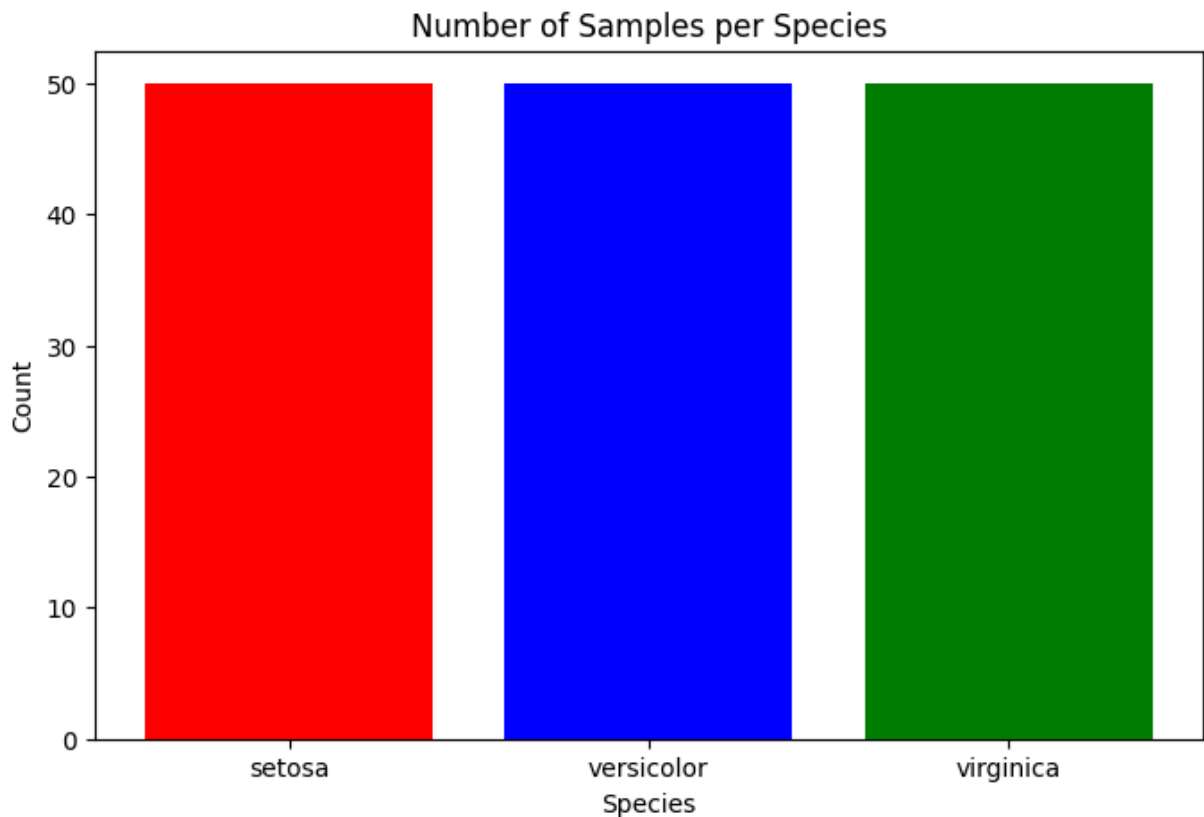
```
plt.xlabel('Species')
```

```
plt.ylabel('Count')
```

```
plt.show()
```

```
print(f"Species distribution: {dict(species_counts)}")
```

```
print("✅ Task 2 completed successfully!")
```



Species distribution: {'setosa': np.int64(50), 'versicolor': np.int64(50), 'virg

✅ Task 2 completed successfully!

Your Analysis and Reflection

****Instructions**:** Complete the analysis below by editing this markdown cell.

My Observations About the Iris Dataset

****Dataset Overview:****

- Number of samples: 150
- Number of features: 4
- Number of classes: 3

****Key Findings from the Visualization:****

1. Setosa appears more separated from the other two species based on sepal length and sepal width.
2. Versicolor and virginica overlap more in the scatter plot, making them harder to distinguish using only these two features.
3. Virginica generally has a larger sepal length than the other species.

****Questions for Further Investigation:****

- Would petal length and petal width separate the three species more clearly?
- Which flower measurement is most useful for predicting the Iris species?

****Reflection:****

During this lab, I learned how to use Python libraries such as Numpy, Pandas and Matplotlib to load, analyze and visualize a dataset. I also learned how scatter plots can help identify patterns and differences between groups of data. This lab helped me better understand how these tools are used in machine learning.

***Note:** This is practice for documenting your machine learning projects professionally.*

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lab helped me better understand how these tools are used in machine learning.

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Lab Summary and Next Steps

What You've Accomplished:

- ✓ Set up your machine learning development environment
- ✓ Imported and used essential Python libraries
- ✓ Loaded and explored your first dataset
- ✓ Created your first data visualization
- ✓ Practiced professional documentation with Markdown
- ✓ Learned about GitHub for project management

Preparation for Module 3:

In the next lab, you'll:

- Learn about different types of machine learning
- Build your first simple classifier
- Understand the complete ML workflow
- Work with more complex datasets

Action Items:

1. **Create your GitHub account** and repository
2. **Upload this completed notebook** to your repository
3. **Experiment** with different visualizations using the Iris dataset
4. **Practice** Markdown formatting in a new notebook

Resources for Continued Learning:

- [Pandas Documentation](#)
- [Matplotlib Gallery](#)
- [GitHub Guides](#)
- [Jupyter Notebook Tips](#)

Great job completing Module 2! You're now equipped with the essential tools for machine learning. 🎉

Module 2 Lab Exercise: Tools Used in Machine Learning

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# Import libraries with standard aliases
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import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets
import warnings
warnings.filterwarnings('ignore') # Hide warning messages for cleaner output

print("✅ All libraries imported successfully!")
print(f"Pandas version: {pd.__version__}")
print(f"NumPy version: {np.__version__}")
```

✅ All libraries imported successfully!
Pandas version: 2.2.3
NumPy version: 2.1.3

✓ Part 2: Loading and Exploring Your First Dataset

We'll use the famous Iris dataset - a classic dataset for beginners. It contains measurements of iris flowers from three different species.

```
# Load a simple dataset (Iris flowers - a classic beginner dataset)
from sklearn.datasets import load_iris
```

```
# Load the data
iris = load_iris()
print("Dataset loaded successfully!")
print(f"Dataset shape: {iris.data.shape}")
print(f"Features: {iris.feature_names}")
print(f"Target classes: {iris.target_names}")
```

```
Dataset loaded successfully!
Dataset shape: (150, 4)
Features: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
Target classes: ['setosa' 'versicolor' 'virginica']
```

```
# Convert to pandas DataFrame for easier handling
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target_names[iris.target]
```

```
# Display first few rows
print("First 5 rows of our dataset:")
print(df.head())
```

```
print("\nDataset info:")
print(df.info())
```

First 5 rows of our dataset:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

```
species
0  setosa
1  setosa
2  setosa
3  setosa
4  setosa
```

Dataset info:

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 150 entries, 0 to 149
```

```
Data columns (total 5 columns):
```

#	Column	Non-Null Count	Dtype
0	sepal length (cm)	150 non-null	float64
1	sepal width (cm)	150 non-null	float64
2	petal length (cm)	150 non-null	float64
3	petal width (cm)	150 non-null	float64
4	species	150 non-null	object

```
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None
```

✓ Part 3: Creating Your First Visualization

Data visualization is crucial in machine learning. Let's create a simple plot to understand our data.

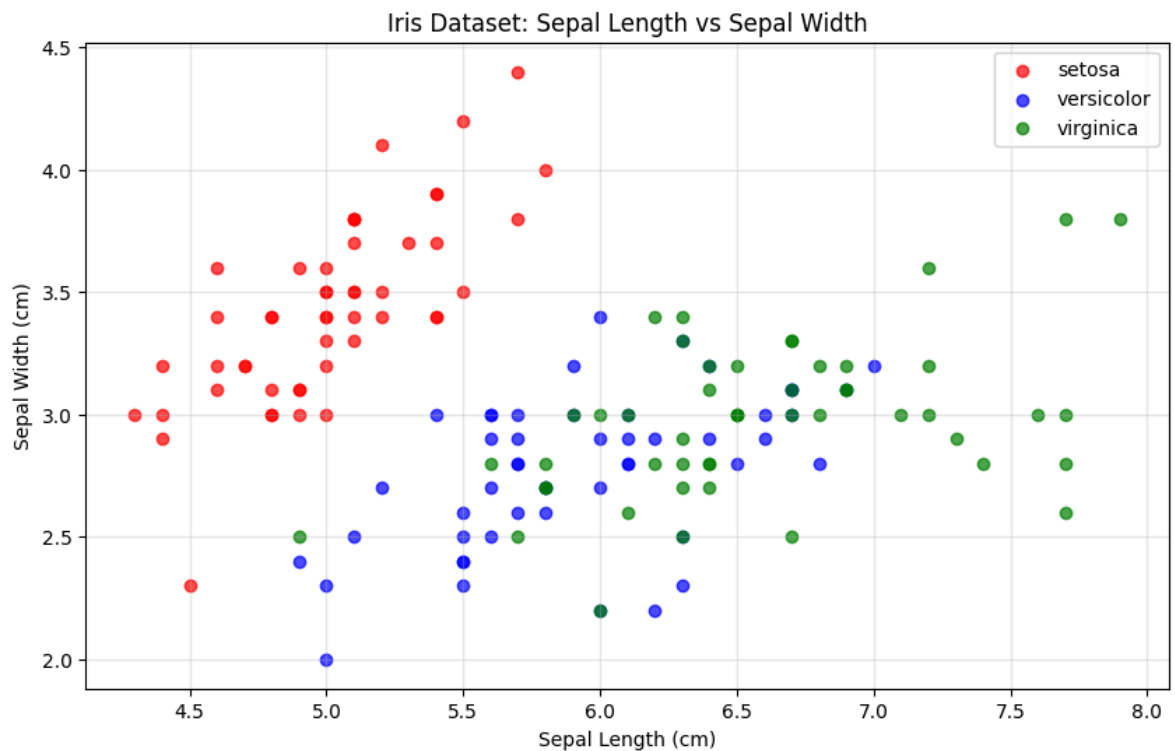
```
# Create a simple scatter plot
plt.figure(figsize=(10, 6))

# Plot sepal length vs sepal width, colored by species
species_colors = {'setosa': 'red', 'versicolor': 'blue', 'virginica': 'green'}

for species in df['species'].unique():
    species_data = df[df['species'] == species]
    plt.scatter(species_data['sepal length (cm)'],
                species_data['sepal width (cm)'],
                c=species_colors[species],
                label=species,
                alpha=0.7)

plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')
plt.title('Iris Dataset: Sepal Length vs Sepal Width')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()

print("🎉 Congratulations! You've created your first data visualization!")
```



🎉 Congratulations! You've created your first data visualization!

✓ Part 4: Practice with Basic Data Operations

Let's practice some basic data analysis operations that you'll use throughout the course.

```
# Basic statistical analysis
print("Basic Statistics for Iris Dataset:")
print("=" * 40)

# Calculate mean values for each species
species_means = df.groupby('species').mean()
print("\nMean values by species:")
print(species_means)

# Count samples per species
species_counts = df['species'].value_counts()
print("\nSamples per species:")
print(species_counts)
```

```
Basic Statistics for Iris Dataset:
=====
```

```

Mean values by species:
      sepal length (cm)  sepal width (cm)  petal length (cm)  \
species
setosa                5.006              3.428              1.462
versicolor           5.936              2.770              4.260
virginica             6.588              2.974              5.552

      petal width (cm)
species
setosa                0.246
versicolor           1.326
virginica             2.026

Samples per species:
species
setosa      50
versicolor  50
virginica   50
Name: count, dtype: int64

```

Part 5: GitHub and Documentation Best Practices

Why GitHub for Machine Learning?

- **Version Control:** Track changes to your code and data
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✓ Assessment: Tool Familiarity Check

Complete the following tasks to demonstrate your understanding of the tools:

```
# Task 1: Create a simple calculation using NumPy
# Calculate the mean and standard deviation of sepal length

sepal_lengths = df['sepal length (cm)']

# Your code here:
mean_sepal_length = np.mean(sepal_lengths)
std_sepal_length = np.std(sepal_lengths)

print(f"Mean sepal length: {mean_sepal_length:.2f} cm")
print(f"Standard deviation: {std_sepal_length:.2f} cm")

# Verification (don't modify)
assert isinstance(mean_sepal_length, (float, np.floating)), "Mean should be a float"
assert isinstance(std_sepal_length, (float, np.floating)), "Std should be a float"
print("✅ Task 1 completed successfully!")
```

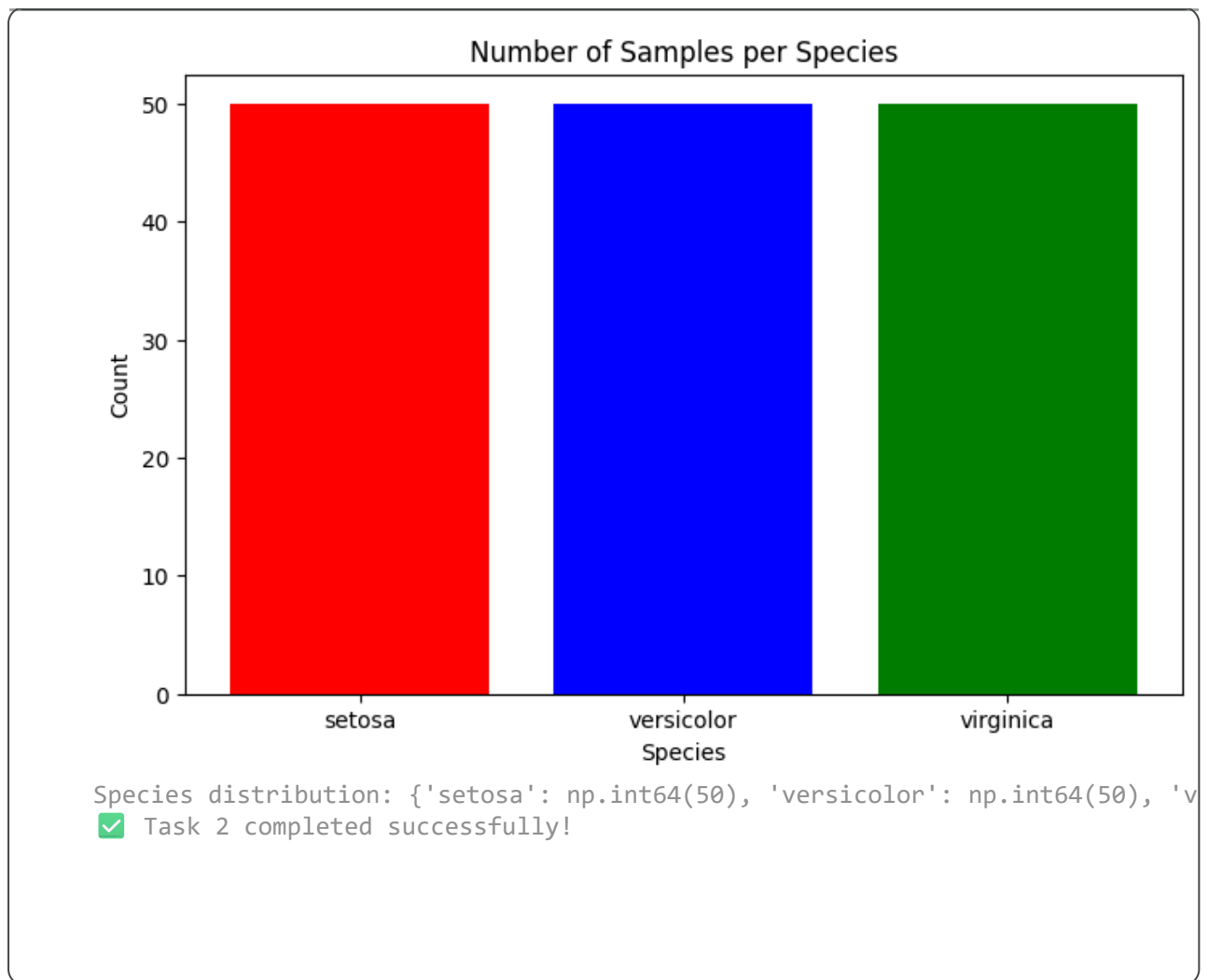
Mean sepal length: 5.84 cm
Standard deviation: 0.83 cm

✅ Task 1 completed successfully!

```
# Task 2: Create a simple bar chart showing species counts
species_counts = df['species'].value_counts()

plt.figure(figsize=(8, 5))
plt.bar(species_counts.index, species_counts.values, color=['red', 'blue', 'green'])
plt.title('Number of Samples per Species')
plt.xlabel('Species')
plt.ylabel('Count')
plt.show()

print(f"Species distribution: {dict(species_counts)}")
print("✅ Task 2 completed successfully!")
```



Your Analysis and Reflection

Instructions: Complete the analysis below by editing this markdown cell.

My Observations About the Iris Dataset

Dataset Overview:

- Number of samples: 150
- Number of features: 4
- Number of classes: 3

Key Findings from the Visualization:

1. The setosa species are more separated, than versicolor and virginia, in the scatter plot.
2. The versicolor and virginica are very similar in their sepal length and width, and that way they overlap in some areas.
3. Setosa have more sepal width than their length.

Questions for Further Investigation:

- Could the petals length and width, separate the species more clearly ?

- Is the scatter plot feature better than a bar chart to compare species ?

Reflection: I learned how pandas can organize a dataset into a DataFrame, making it easier to compare three different flower species. The two charts, reveal the pattern that are difficult to recognize by looking only at numerical information

Note: This is practice for documenting your machine learning projects professionally.

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- ✓ Set up your machine learning development environment
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In the next lab, you'll:

- Learn about different types of machine learning
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0	5.1	3.5	1.4	0.0	setosa
1	4.9	3.0	1.4	0.0	setosa
2	4.7	3.2	1.3	0.0	setosa
3	4.6	3.1	1.5	0.0	setosa
4	5.0	3.6	1.4	0.0	setosa

```
species
0  setosa
1  setosa
2  setosa
3  setosa
4  setosa
```

```
Dataset info:
<class 'pandas.core.frame.DataFrame'>
```

```

RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   sepal length (cm)      150 non-null    float64
1   sepal width (cm)       150 non-null    float64
2   petal length (cm)      150 non-null    float64
3   petal width (cm)       150 non-null    float64
4   species                150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None

```

✓ Part 3: Creating Your First Visualization

Data visualization is crucial in machine learning. Let's create a simple plot to understand our data.

```

# Create a simple scatter plot
plt.figure(figsize=(10, 6))

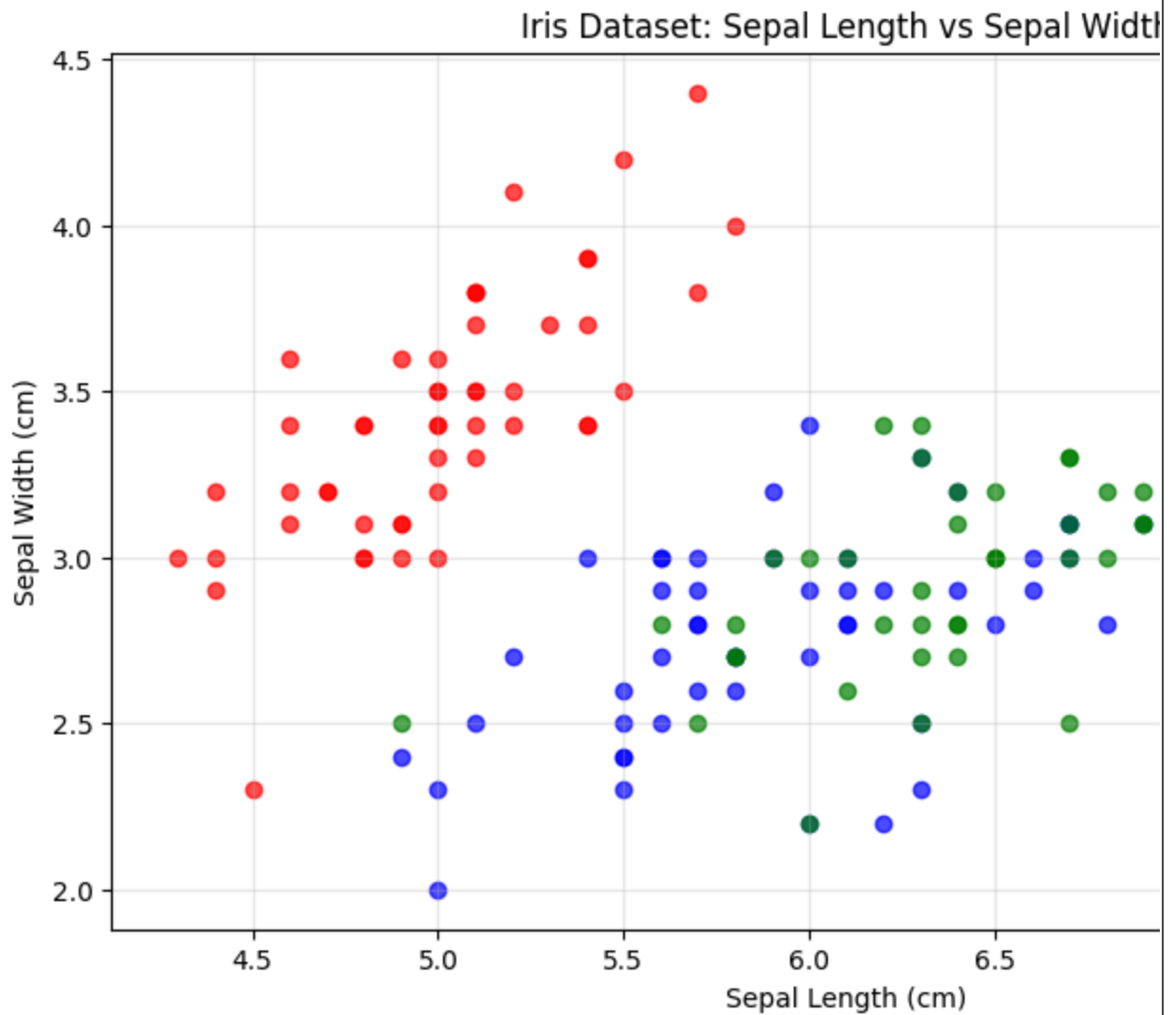
# Plot sepal length vs sepal width, colored by species
species_colors = {'setosa': 'red', 'versicolor': 'blue', 'virginica': 'g'}

for species in df['species'].unique():
    species_data = df[df['species'] == species]
    plt.scatter(species_data['sepal length (cm)'],
                species_data['sepal width (cm)'],
                c=species_colors[species],
                label=species,
                alpha=0.7)

plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')
plt.title('Iris Dataset: Sepal Length vs Sepal Width')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()

print("🎉 Congratulations! You've created your first data visualization!")

```



Congratulations! You've created your first data visualization!

✓ Part 4: Practice with Basic Data Operations

Let's practice some basic data analysis operations that you'll use throughout the course.

```
# Basic statistical analysis
print("Basic Statistics for Iris Dataset:")
print("=" * 40)

# Calculate mean values for each species
species_means = df.groupby('species').mean()
print("\nMean values by species:")
print(species_means)

# Count samples per species
species_counts = df['species'].value_counts()
print("\nSamples per species:")
print(species_counts)
```

Basic Statistics for Iris Dataset:

=====

Mean values by species:

	sepal length (cm)	sepal width (cm)	petal length (cm)	\
species				
setosa	5.006	3.428	1.462	
versicolor	5.936	2.770	4.260	
virginica	6.588	2.974	5.552	

	petal width (cm)
species	
setosa	0.246
versicolor	1.326
virginica	2.026

Samples per species:

species	
setosa	50
versicolor	50
virginica	50

Name: count, dtype: int64

Part 5: GitHub and Documentation Best Practices

Why GitHub for Machine Learning?

- **Version Control:** Track changes to your code and data
- **Collaboration:** Work with others on projects
- **Portfolio:** Showcase your work to potential employers
- **Backup:** Never lose your work

Basic GitHub Workflow:

1. **Create Repository:** A folder for your project
2. **Clone/Download:** Get the project on your computer
3. **Add Files:** Put your notebooks and data
4. **Commit:** Save a snapshot of your changes
5. **Push:** Upload changes to GitHub

For This Course:

- Create a repository named "ITAI-1371-ML-Labs"
- Upload each lab notebook as you complete it
- Include a README.md file describing your projects

Action Item: After this lab, create your GitHub account and repository.

✓ Assessment: Tool Familiarity Check

Complete the following tasks to demonstrate your understanding of the tools:

```
# Task 1: Create a simple calculation using NumPy
# Calculate the mean and standard deviation of sepal length

sepal_lengths = df['sepal length (cm)']

# Your code here:
mean_sepal_length = np.mean(sepal_lengths)
std_sepal_length = np.std(sepal_lengths)

print(f"Mean sepal length: {mean_sepal_length:.2f} cm")
print(f"Standard deviation: {std_sepal_length:.2f} cm")

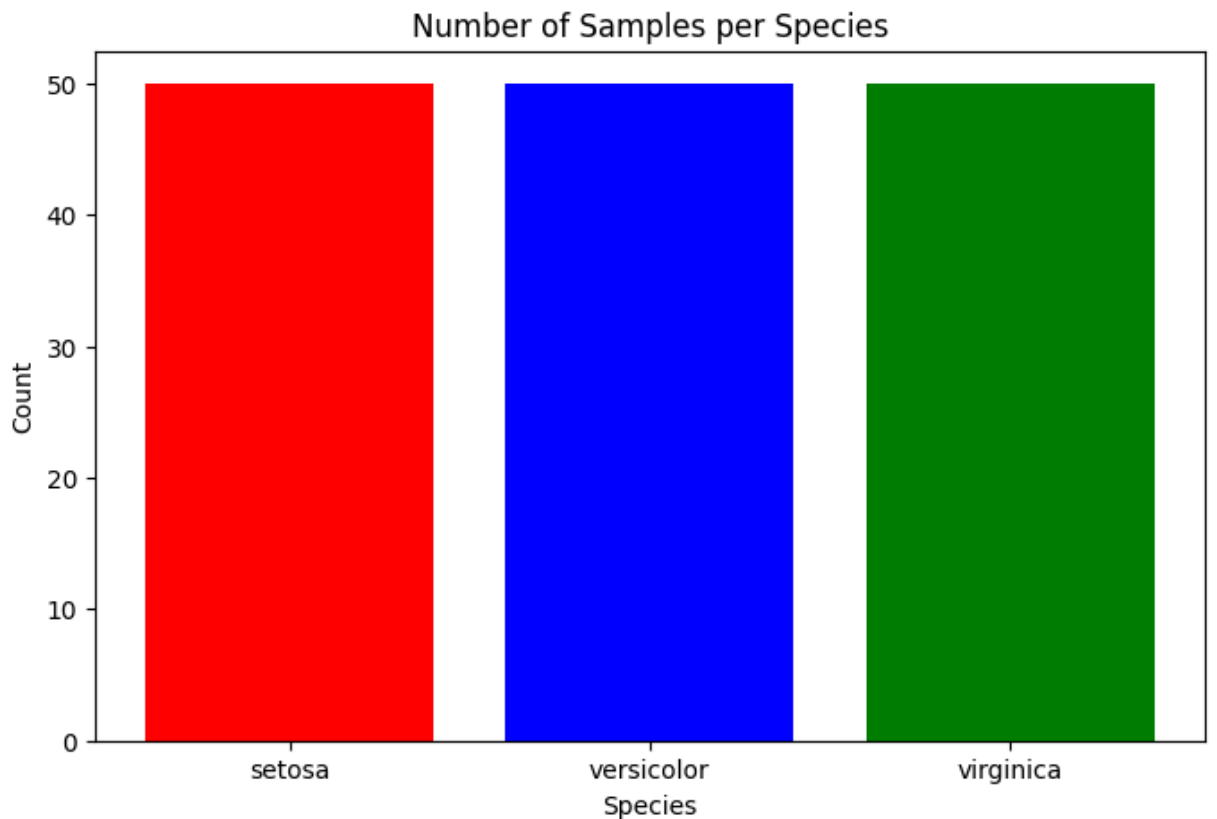
# Verification (don't modify)
assert isinstance(mean_sepal_length, (float, np.floating)), "Mean should be float"
assert isinstance(std_sepal_length, (float, np.floating)), "Std should be float"
print("✅ Task 1 completed successfully!")
```

```
Mean sepal length: 5.84 cm
Standard deviation: 0.83 cm
✅ Task 1 completed successfully!
```

```
# Task 2: Create a simple bar chart showing species counts
species_counts = df['species'].value_counts()

plt.figure(figsize=(8, 5))
plt.bar(species_counts.index, species_counts.values, color=['red', 'blue'])
plt.title('Number of Samples per Species')
plt.xlabel('Species')
plt.ylabel('Count')
plt.show()

print(f"Species distribution: {dict(species_counts)}")
print("✅ Task 2 completed successfully!")
```



Species distribution: {'setosa': np.int64(50), 'versicolor': np.int64(50)}
✅ Task 2 completed successfully!

Your Analysis and Reflection

Instructions: Complete the analysis below by editing this markdown cell.

My Observations About the Iris Dataset

Dataset Overview:

- Number of samples: 150
- Number of features: 4
- Number of classes: 3

Key Findings from the Visualization:

1. The 'Setosa' class generally has a shorter sepal than the 'Versicolor' and 'Virginica' classes.
2. The 'Versicolor' and 'Virginica' generally have larger width than the 'Setosa' class.
3. The 'Virginica' class is generally the largest, in both sepal length and width of the 3 classes.

Questions for Further Investigation:

- I'd like to see this with a larger sample size for each class.

Reflection: *In 2-3 sentences, describe what you learned about using these tools.* There does not seem to be a lot happening in the code because so much of it is abstracted away. Even so, there is a lot of code making it a little overwhelming to try to understand everything.

Note: This is practice for documenting your machine learning projects professionally.

Lab Summary and Next Steps

What You've Accomplished:

- ✓ Set up your machine learning development environment
- ✓ Imported and used essential Python libraries
- ✓ Loaded and explored your first dataset
- ✓ Created your first data visualization
- ✓ Practiced professional documentation with Markdown
- ✓ Learned about GitHub for project management

Preparation for Module 3:

In the next lab, you'll:

- Learn about different types of machine learning
- Build your first simple classifier
- Understand the complete ML workflow
- Work with more complex datasets

Action Items:

1. **Create your GitHub account** and repository
2. **Upload this completed notebook** to your repository
3. **Experiment** with different visualizations using the Iris dataset
4. **Practice** Markdown formatting in a new notebook

Resources for Continued Learning:

- [Pandas Documentation](#)
- [Matplotlib Gallery](#)
- [GitHub Guides](#)

- [Jupyter Notebook Tips](#)